

Introduction to Web Programming project documentation

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Introduction

Weather app was done for the project. In the weather app users can search for the temperature of location. Location can be searched by latitude and longitude or location name by reverse geolocation and geolocation respectively. Whenever the user searches location by latitude and longitude application automatically fills in location name and vice versa with location name.

App shows in two charts temperature for next 24 hours and next 7 days. Temperature data in chart for 7 next days are fetched from MET Norway API and data is only for Northern Europe if data is fetched outside this then chart is not updated or doesn't show up. In the temperature chart for 24 hours data are fetched from Google Weather API and Open-Meteo API. Current temperature, highest and lowest temperature of current day, wind speed, UV index, weather description and weather icon are also shown. Temperatures are shown in Celsius, but users can convert to Fahrenheit or Kelvin. Charts can be exported as SVG files. Locations can be saved as favorites which are then shown in favorites list. Users can delete favorites from top of the list, bottom of the list or delete all favorites.

In the end of the page users can go to a new page that has generated QR-code and link for Zoom Earth website that has location already inputted. In the same page there is also a rain or snow animation whether there is rain or snow in the current search location. Animation is faster when there is more rain or snow. App turns in dark mode when app is used after sunset and before sunrise according to the current location (latitude: 61.05497219999999 and longitude: 28.1896039, Lappeenranta). Dark mode is inspired by Google's Material Design as background isn't fully black. If the application is used outside of the dark mode window background reflects the temperature of the searched location.

Tools

Tools used in making the project are from course material, weekly tasks from previous weeks, Frappe charts documentation and Google searching. Application's dark mode is influenced by Google's Material Design dark theme principles. Bootstrap icons are from Bootstrap documentation.

This website below was used as a training to get a right regex code.

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/String/match.

Background colors in different temperatures are defined by the image below.

°F	°C	Intuitive interpretation
104	40	Clothing catches fire
95	35	Damn hot
86	30	Quite hot
77	25	Warm
68	20	Nice
59	15	Cool
50	10	Chilly
41	5	Cold
32	0	Freezing
23	-5	Quite cold
14	-10	Really cold
5	-15	Damn cold
-4	-20	Really damn cold

The code for QR code generation is from here: <https://www.youtube.com/watch?v=X6MFUagtKiQ>.

AI usage:

GitHub CoPilot gave autofill recommendations for using localStorage as storing data as global variable to use between two pages. Autofill also gave recommendations for using window.href.location command for switching between pages and for-loop structure to check if location to be added into favorites list is already there.